Sonakshi Godavarthy

AI / Machine Learning Engineer | NLP & Information Extraction | LLMs & Multimodal AI

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portfolio-site-sonakshi.vercel.app LinkedIn GitHub

PROFILE

AI/ML engineer with experience in building and evaluating NLP, generative AI, multimodal learning, and data-driven applications. Skilled in Python, PyTorch, transformer models, information extraction, retrieval, and model evaluation. Delivered research and applied projects across technical document processing, synthetic data generation, and machine learning.

TECHNICAL SKILLS

Programming, Machine Learning & Data Science
Python, PyTorch, Scikit-learn, CNNs, Logistic Regression, Decision Trees, Representation Learning, Federated Learning (FedAvg, FedProx, FedNova), Concept Drift, Non-IID Learning, Hyperparameter Tuning, Pandas, NumPy

Multimodal AI, Synthetic Data & Explainability
Multimodal Representation Learning, Synthetic Data Generation, Synthetic Data Vault (SDV), PyTorch Diffusion Models, Denoising MLPs, RLHF, Reward Modelling, REINFORCE, Bradley–Terry Reward Models, Counterfactual Explanations

Model Evaluation & Data Analysis
ROUGE, Precision@K, Statistical Validation, Schema Checks, Distribution Comparison, Outlier Detection, Error Analysis, Heatmaps, t-SNE

Generative AI, NLP & Information Extraction
LLMs, Hugging Face Transformers, GPT-4, Qwen, Mistral, DeBERTa, GLiNER, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Fine-Tuning, Semantic Retrieval, Text Summarisation, Document Understanding, Technical Document Understanding, Information Extraction, Named Entity Recognition, DistilBART-CNN, all-MiniLM-L6-v2

Data Engineering, Streaming & Databases
SQL, ETL, Apache Kafka, Apache Flink, Informatica Cloud (IICS), Database Connectors, PostgreSQL, MySQL, Oracle, Snowflake, Neo4j

Software Development, Deployment & Cloud
Django, React, FastAPI, REST APIs, Docker, Streamlit, Figma, Linux, Git, GitHub, Claude Code (AI-assisted coding), AWS, Azure

EDUCATION

M.Sc. Data Science Oct 2023 – Dec 2026 | Hamburg, Germany
Hamburg University of Technology (TUHH)
Current average: 1.9 (German scale; 1.0 is best)

B.Tech. Computer Science & Engineering Jun 2018 – May 2022 | Hyderabad, India
CMR College of Engineering & Technology
GPA: 8.87/10 (1.6/5.0, German scale)

PROFESSIONAL EXPERIENCE

Deutsches Zentrum für Luft- und Raumfahrt (DLR) Apr 2026 – Present | Hamburg, Germany
Master's Thesis Student

- Researching multimodal machine-learning methods for representing complex Airbus A320 maintenance documentation containing text, tables, equations, flowcharts, and technical diagrams.
- Developing pipelines that transform heterogeneous technical information into structured representations for machine reasoning and grounded retrieval.
- Investigating representation, retrieval, and reasoning strategies for combining structured and unstructured technical knowledge.

Lufthansa Technical Training GmbH Nov 2025 – Mar 2026
Intern - eLearning and Artificial Intelligence Hamburg, Germany

- Built functional prototypes of a RAG-based document-processing pipeline supporting content summarisation, question generation, information retrieval and learner assistance.
- Designed solution architectures using DistilBART-CNN for summarisation, all-MiniLM-L6-v2 for semantic retrieval and GPT-4 for question generation and learner support.
- Evaluated model and retrieval performance using ROUGE and Precision@K.

Fraunhofer Institute for Mechatronic Systems Design Jan 2025 – Aug 2025
Research Assistant (GenAI Incubator) Paderborn, Germany

- Developed and compared synthetic tabular-data generation methods using GPT-4 Turbo, Synthetic Data Vault (SDV), custom PyTorch diffusion models, denoising MLPs and rule-based techniques, generating 15,000+ synthetic records across experiments.
- Integrated the methods into an automated pipeline that analysed dataset columns, generated structured synthetic data and created component descriptions.

- Validated synthetic-data quality using schema checks, statistical distribution comparisons and outlier detection, with 98% of generated records passing the defined validation checks.
- Designed the interface in Figma and implemented it as a React web application, allowing users to upload datasets, select a generation model and generate additional synthetic data interactively.

Institute of Data Science Foundations (TUHH) Feb 2025 – Jul 2025 | Hamburg, Germany
Student Assistant
- Analysed EuXFEL laboratory data to investigate why particle-beam simulations from physics-based simulation software did not always match the actual experimental measurements.
- Built a Python workflow to compare 189 real measurements with 189 API-generated simulation results using error analysis, heatmaps and t-SNE; 88.9% of the cases met the defined error threshold between simulated and measured beam-quality values.
- Investigated the 21 high-error cases and found that 86% occurred on the same measurement date, indicating possible experiment-specific or data-quality issues.

Informatica Jul 2022 – Sep 2023 | Bangalore, India
Associate Software Engineer - Global Customer Support
- Resolved complex Informatica Cloud integration and connector issues, achieving a 95% customer satisfaction rate.
- Optimised ETL workflows (Informatica Cloud) with various database connectors, improving data pipeline efficiency and reliability.
- Developed successful technical workarounds for approximately 90% of assigned support cases and authored reusable knowledge articles to reduce repeated investigation of recurring cases.

Informatica Aug 2021 – Jun 2022 | Bangalore, India
Intern - Global Customer Support
- Completed technical training in Informatica Intelligent Cloud Services (IICS) and worked with flat-file, Oracle, MySQL, PostgreSQL and Snowflake connectors for data integration and transformation.
- Collaborated with senior engineers to prioritise and resolve customer issues, enhancing issue response times by 20%.

PROJECTS

Applied Machine Learning & Human-Centered AI Apr 2025 – Sep 2025
- Developed a Django-based AutoML application for CSV upload, data visualisation, model training and hyperparameter tuning.
- Built a TF-IDF and Logistic Regression sentiment classifier for 50,000 IMDB reviews, achieving 89.8% test accuracy, with active-learning strategies.
- Created an explainable AI interface using decision trees, sparse Logistic Regression and counterfactual explanations.
- Developed a movie recommender using Truncated SVD on MovieLens 32M, generating personalised Top-5 recommendations.
- Implemented a PyTorch RLHF pipeline using a CNN policy, REINFORCE, human preferences and a Bradley–Terry reward model.

Exploiting Tree-Based Topologies and Aggregation Functions Against Concept Drift in Hierarchical Federated Networks Oct 2024 – Jul 2025
- Investigated how Hierarchical Federated Learning systems maintain accuracy when client data changes over time (concept drift) and some clients become unavailable.
- Developed a PyTorch-based framework using a CNN on non-IID MNIST data and evaluated FedAvg, FedProx, FedNova and hybrid aggregation under synchronous, gradual and abrupt concept drift with 40% client dropout.
- Found that FedProx and FedNova adapted more effectively than FedAvg, while hybrid aggregation achieved the strongest recovery in client-dropout scenarios.

YouTube Transcript Summarisation Aug 2024 – Aug 2024
- Developed an automated video summarisation pipeline using fine-tuned DistilBART-CNN models to generate concise summaries from YouTube transcripts.
- Integrated transcript extraction and NLP preprocessing into an automated summarisation pipeline, reducing content review time by 60%.

Real-time Traffic Monitoring Apr 2024 – Aug 2024
- Developed a scalable streaming data pipeline to process sensor data in real-time using Kafka and Flink, enabling structured fleet monitoring dashboards.
- Applied custom operators to calculate taxi speed, distance, and performance metrics.
- Created a web interface using Streamlit to display real-time fleet information and alerts.

LANGUAGES

English — Fluent | **German** — A2, currently learning | **Hindi** — Fluent | **Telugu** — Fluent

SCHOLARSHIP

Merit Scholarship for International Students Aug 2025
Hamburg University of Technology (TUHH)
- Awarded for academic excellence